

Compare between Wireless BDA and Fiber ORU

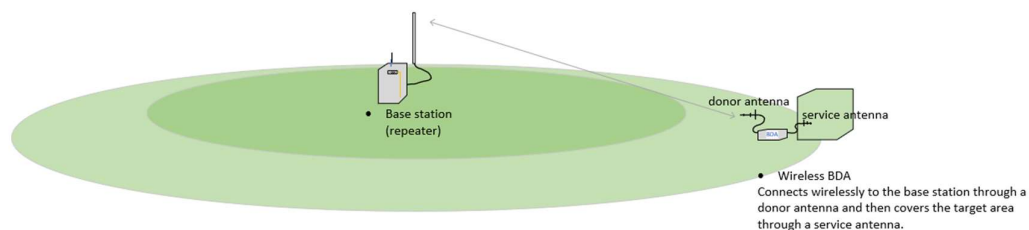
Fiber ORU (Optical Remote Unit) has increasingly replaced Wireless BDA (Bi-Directional Amplifier) in many scenarios. This shift is largely due to the widespread adoption of fiber optics and the growing demand from users for broader coverage areas and higher signal quality.

The Fiber ORU offers superior performance in maintaining signal integrity over long distances, providing more consistent and reliable coverage compared to Wireless BDA systems, which are more susceptible to signal degradation and interference.

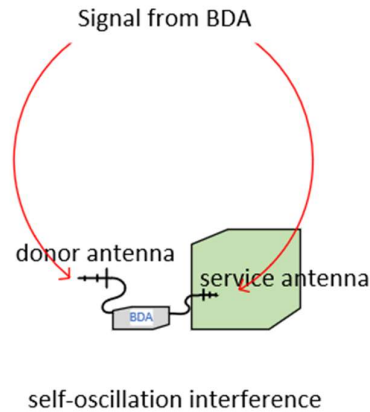
The document compares the working principles and the impact on system performance between **Wireless BDA (Bi-Directional Amplifier)** and **Fiber ORU**.

Wireless BDA:

- **How it Works:** Connects wirelessly to the base station through a donor antenna and then covers the target area through a service antenna.



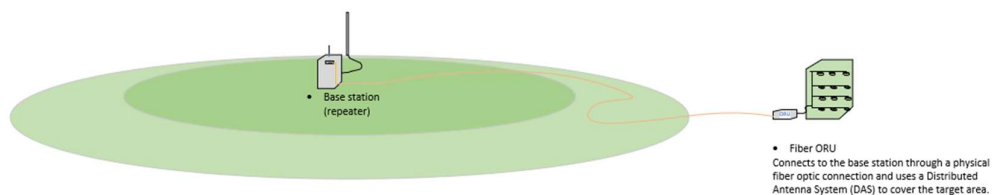
- **Advantages:**
 - Flexible installation, as it does not require a physical connection to the base station.
 - Can extend the signal coverage within the base station's range.
- **Disadvantages:**
 - Requires a high-quality signal from the donor antenna, so coverage extension is limited by the base station's signal quality.
 - Requires high isolation between the donor and service antennas to prevent self-oscillation.
 - The amplification factor is limited by the received signal quality and potential self-oscillation issues, limiting the coverage area.
 - Wireless BDAs can generate noise and self-oscillation, which can reduce the overall coverage area of the base station by affecting its sensitivity.



The principle behind signal self-oscillation interference in a Bi-Directional Amplifier (BDA) occurs due to the BDA's operation on the same frequency for both the uplink and downlink. The BDA amplifies signals received by the donor antenna and retransmits them through the service antenna. However, if the amplified signal from the service antenna is picked up again by the donor antenna due to insufficient spatial isolation between them, the BDA will re-amplify its own transmitted signal. This BDA amplification creates a feedback loop, leading to a phenomenon known as self-oscillation. This self-oscillation can cause severe interference, degrading the overall performance of the communication system and potentially disrupting the signal environment.

Fiber ORU:

- **How it Works:** Connects to the base station through a physical fiber optic connection and uses a Distributed Antenna System (DAS) to cover the target area.



- **Advantages:**
 - High signal quality transmission and flexible extension where fiber resources are available.
 - No issues with self-oscillation.
 - Direct fiber connection ensures high signal quality, allowing for better amplification and larger area coverage through the DAS system.
 - High isolation between the receiving and transmitting parts prevents self-

oscillation and interference, allowing for more extensive use of BDAs.

- **Disadvantages:**

- Dependent on the availability of fiber resources for extension.

Wireless BDA	Fiber ORU
Prone to self-oscillation, deteriorating the electromagnetic environment in the coverage area.	No self-oscillation, resulting in higher signal
Amplification factor is affected by the received signal quality, limiting DAS system expansion.	Can drive a larger area DAS system.
Can interfere with the base station, reducing its sensitivity, with limited expansion of BDA units.	Does not interfere with the base station, allowing for more BDAs to expand the coverage area.
Flexible installation; can be installed independently within the base station's coverage area.	Requires fiber resources; can extend coverage beyond the base station's range if fiber is available.